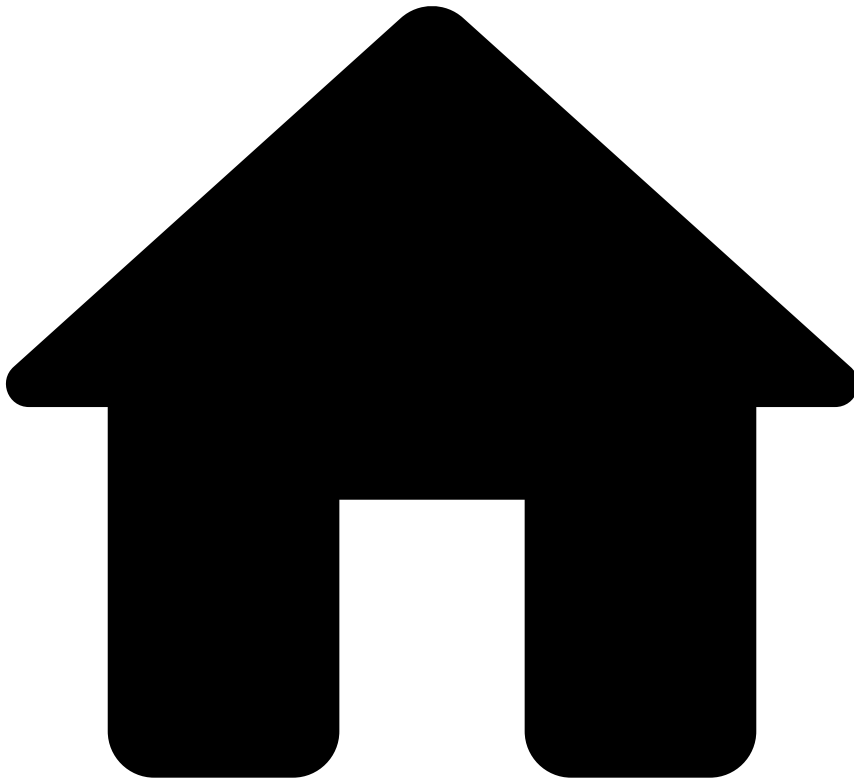


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- [Core Concepts](#)
- User-Defined Functions

On this page

User-Defined Functions

Was this page helpful? 

Pulse comes with a set of User-Defined Functions (UDFs) that can be used in PQL expressions to implement richer logic than what's available from the basic built-in functions. This page provides an overview of the available UDFs.

To learn about how to use UDFs, see [Function Calls in PQL](#).

Working with strings

`DataConversionTools` [class](#) 

Package: `com.feedzai.common.udfs`

Modifier and Type	Function	Description
static Integer	stringToInt(String s)	Converts a String into an Integer. If the given s is null or the value exceeds Integer.MAX_VALUE or Integer.MIN_VALUE , then null is returned.
static Long	stringToLong(String s)	Converts a String into a Long. If the given s is null or the value exceeds Long.MAX_VALUE or Long.MIN_VALUE , the null is returned.
static Double	stringToDouble(String s)	Converts a String into a Double. If the given s is null or the value exceeds Double.MAX_VALUE or Double.MIN_VALUE , then null is returned.
static Float	stringToFloat(String s)	Converts a String into a Float. If the given s is null or the value exceeds Float.MAX_VALUE or Float.MIN_VALUE , then null is returned.
static String	integerToString(Integer i)	Converts an Integer into a String. If the given Integer is null, null is returned.
static String	longToString(Long l)	Converts a Long into a String. If the given Long is null, null is returned.
static String	doubleToString(Double d)	Converts a Double into a String. If the given Double is null, null is returned.
static String	floatToString(Float f)	Converts a Float into a String. If the given Float is null, null is returned.

Strings class (fraud)🔗📄

Package: `com.feedzai.pulse.fraud.udf`

Modifier and Type	Function	Description
static Integer	levenshteinDistance(String str1, String str2)	Computes the Levenshtein distance (edit distance) of two strings.
static Boolean	matches(String input, String like)	Returns true if the input matches the 'like' pattern, using case-sensitive matching.
static Boolean	matchesIgnoreCase(String input, String like)	Returns true if the input matches the 'like' pattern, using case-insensitive matching.
static String	strcat(Object obj1, Object obj2)	Concatenates two strings.

Strings class (cloud)🔗📄

Package: `com.feedzai.cloud.udf`

Modifier and Type	Function	Description
Integer	tokenCount(str: String, delimiter: String)	Returns the number of tokens in a given string, separated by the defined delimiter. For example, <code>tokenCount("Number of substrings", " ") = 3</code> .
Double	twoGramSimilarity(str1: String, str2: String)	Computes the 2-gram similarity between two strings. The returned value is a double in a range from 0.0 to 1.0, where 0.0 represents the minimum similarity and 1.0 represents maximum similarity. The 2-gram similarity between strings is computed after white space removal. If the given strings are null they will be considered empty strings. For more details see the documentation of NGramDistance .
String	normalize(str: String)	Removes accents and converts the string to lower case. If the input string is null, an empty string is returned.
String	normalizeDelWhitespace(str: String)	Removes the accents, deletes white spaces and converts the string to lower case. If the input string is null, an empty string is returned.
String	normalizeToAlphanumeric(str: String)	Normalizes the string to contain only ascii alphanumeric symbols. Note that this will strip accents, so "Ãƒ" will become "a".

Modifier and Type	Function	Description
Boolean	isSubstring(str: String, sub: String)	Returns true if a string contains a given substring in a case insensitive manner.
Boolean	isAllCaps(str: String)	Returns true if the input string is all upper case.
Boolean	isAllNonWhitespaceCaps(str: String)	Returns true if the input string is all upper case. Ignores white spaces.
Integer	strDistance(str1: String, str2: String)	Returns the Levenshtein-distance between two strings.
String	extractDigits(str: String)	Returns the digits from a string.
String	dropDigits(str: String)	Returns the string after dropping the digits from it.
Boolean	isDigits(str: String)	Returns true if the string consists only of digits.
Integer	length(str: String)	Returns the length of the string, or 0 if the string is null.
Integer	countChars(str: String, chars: String)	Counts the number of characters of "chars" in "str", or 0 if any of the input strings is null.
Integer	longestSameLetterSubstringLength(str: String)	Returns the length of the longest substring that is a repetition of the same letter.
Integer	longestIncreasingSubstringLength(str: String)	The length of the longest substring that is strictly increasing by one unit, e.g. "defgh".
Boolean	containsCyrillic(str: String)	Returns true if a string contains Cyrillic characters. If the string is empty, returns false.
Boolean	regexFind(regex: String, str: String)	Returns true if a string matches a given regex pattern. If any of the parameters is empty, the function returns false.
Double	keyboardDistance(str: String, step: Int)	Returns the average keyboard distance between every character in a string and the character that precedes it by a given step.
Double	numRatio(str: String)	Returns the ratio of digits in the input string.
Double	vowelRatio(str: String)	Returns the ratio of vowels (a, e, i, o, u, Ð°, Ð¾, Ñ, Ñ, Ñ) in the input string.
Double	distinctCharsRatio(str: String)	Returns the ratio of distinct characters in the input string.

StringTools class

Package: `com.feedzai.commons.udfs`

Modifier and Type	Function	Description
static Integer	compareTo(String f, String s)	Lexicographically compares string f with string s.
static String	concat(String...s)	Joins the elements of the provided array s into a single string object.
static Boolean	contains(String f, String s)	Checks if string f contains string s.
static String	deleteWhitespace(String s)	Deletes all whitespaces from a string as defined by Character.isWhitespace(char).
static Double	getDiceIndex(String f, String s)	Returns the Š,ensenšDice index for a given pair of strings.
static Integer	getLevenshteinDistance(String s, String t)	Returns the Levenshtein distance between two strings.
static Double	getNormalizedLevenshteinDistance(String s, String t)	Returns the Levenshtein distance between two strings, divided by the length of the longest string.
static Integer	length(String text)	Returns the length of a string.
static String	lowerCase(String s)	Converts string s to lower case as per String.toLowerCase() .

Modifier and Type	Function	Description
static Boolean	regexMatches(String regex, String text)	Checks if a string matches a given regular expression.
static String	replaceAll(String regex, String text, String replacement)	In the text, this function replaces every subsequence that matches the regular expression with the defined replacement.
static String	replaceFirst(String regex, String text, String replacement)	In the text, this function replaces the first subsequence that matches the regular expression with the defined replacement.
static java.util.List<String>	regexSplit(String regex, String text)	Splits a given text according to a given regex.
static String	replace(String text, String find, String replace)	Replaces all occurrences of a string (find) within a text.
static java.util.List<String>	split(String s)	Splits the provided text into an array, using whitespace as the separator.
static java.util.List<String>	split(String s, String sep)	Splits the provided text into an array, using whitespace as the separator.
static String	stripAccents(String s)	Removes accents from a given string.
static String	substring(String text, Integer begin)	Gets a substring from the specified string, avoiding exceptions.
static String	substring(String text, Integer begin, Integer end)	Gets a substring from the specified string, avoiding exceptions.
static String	trim(String s)	Deletes all start and end whitespaces from a string as String.trim().
static String	trimOrNull(String s)	Deletes all start and end whitespaces from a string as String.trim().
static String	upperCase(String s)	Converts a string to upper case as per String.toUpperCase().

Parse class

Package: `com.feedzai.cloud.udf`

Modifier and Type	Function	Description
Integer	asInt(str: String)	Convert the given string to an integer.
Long	asLong(str: String)	Convert the given string to a Long.
Double	asDouble(str: String)	Convert the given string to a Double.
Boolean	asBoolean(str: String)	Convert the given string to a Boolean.

MoreStrings class

Package: `com.feedzai.ds.udf`

Modifier and Type	Function	Description
static Integer	countDigits(String s)	Counts the number of digits in a string.
static Integer	countLetters(String s)	Counts the number of letters in a string.
static Integer	countLowerCase(String s)	Counts the number of lower case letters in a string.
static Integer	countUpperCase(String s)	Counts the number of upper case letters in a string.
static String	replaceAll(String text, String regex, String replacement)	Replaces the regex occurrence in the text with the replacement.

Working with location and addresses

Address class

Package: `com.feedzai.cloud.udf`

Modifier and Type	Function	Description
String	joinAddressFields(name: String, line1: String, line2: String, zip: String, city: String, region: String, country: String)	Joins all the address fields into a single string, including the person's name. Null fields are treated as empty strings.
String	joinAddressFields(line1: String, line2: String, zip: String, city: String, region: String, country: String)	Joins all the address fields into a single string, excluding the person's name. Null fields are treated as empty strings.
String	extractNumericFields(str: String)	Returns the space separated numeric fields from the address, or an empty string if no numeric fields exist.
String	extractNumericFields(line1: String, line2: String, zip: String)	Returns the numeric fields from an address broken down into two lines and zip code. Returns an empty string if no numeric fields exist.
Boolean	isMismatch(fields1: String, fields2: String)	Returns true if the fields are not empty and different.
String	basicNormalization(address: String)	Normalizes the address to alphanumeric characters.
String	basicNormalization(name: String, line1: String, line2: String, zip: String, city: String, region: String, country: String)	Concatenates the parameters and normalizes the string to alphanumeric characters.
String	addressNormalizationReplacement(address: String, AddressNormalizationReplacements: Seq[(String, String)])	Applies replacement rules to a basic-normalized address, replacing each source token by its configured abbreviation/value.
String	advancedNormalization(address: String)	Replaces address suffixes with their abbreviation (e. g.: street to str), and normalizes the string to alphanumeric characters.
String	aggressiveNormalization(address: String)	Executes the advancedNormalization() function, then moves the numbers in the address to the front, and removes arithmetic progression substrings.
String	concatNormalizedFields(fields: String*)	Normalizes and concatenates the input strings into a single string. Removes all whitespaces and non-word characters, including applying Strings.normalizeToAlphanumeric().
String	concatNormalizedFieldsList(fields: JList[String])	Normalizes and concatenates the input strings into a single string. Removes all whitespaces and non-word characters, including applying Strings.normalizeToAlphanumeric().
String	normalizeUsaAddress(zip: String, line1: String)	Normalizes a US address to the most essential information of the address: the first digits of the zip code up to 4 digits, and the first characters of the normalized address line (10-14 characters, depending on the length of the zip), concatenated in <zip>_<normalizedaddress> format, and cut to 15 characters.
String	aggressiveNormalizations(input: String*)	Applies aggressive normalization after concatenating the nullable sequence of strings given as input.

AddressUtils class

Package: `com.feedzai.ds.udf`

Modifier and Type	Function	Description
static Integer	countAddressSuffixes(String input)	Counts the suffixes in an address. Examples for suffixes: <i>av</i> , <i>ave</i> , <i>avenue</i> , <i>st</i> , <i>str</i> , <i>street</i> .

Modifier and Type	Function	Description
static Boolean	isEqualNormalized(String a, String b)	Returns true if the two input strings are equal after making both lower case, removing the address suffixes, ordinals, special characters and spaces.
static String	normalizeZipUK(String input)	Normalizes a UK zip code by removing non-alphanumeric characters and, when possible, keeping only the outward-code portion.

Country class

Package: `com.feedzai.pulse.fraud.udf`

Modifier and Type	Function	Description
staticString	alpha2ToAlpha3(String country)	Converts an alpha2 country code to alpha3 .
staticInteger	alpha2ToIso(String country)	Converts an alpha2 country code to numeric.
staticString	alpha3ToAlpha2(String country)	Converts an alpha3 country code to alpha 2.
staticInteger	alpha3ToIso(String country)	Converts an alpha3 country code to numeric.
staticString	getContinentFromAlpha2(String country)	Gets the continent from a country by its alpha2 code.
staticString	getContinentFromAlpha3(String country)	Gets the continent from a country by its alpha3 code.
staticString	getContinentFromIso(Integer country)	Gets the continent from a country by its iso numeric code.
staticString	isoToAlpha2(Integer country)	Converts a numeric iso country code to alpha 2.
staticString	isoToAlpha3(Integer country)	Converts a numeric iso country code to alpha 3.

DistanceUtils class

Package: `com.feedzai.common.udfs.distance`

Modifier and Type	Function	Description
static Double	distance(String curZipCode, String prevZipCode, String curCityName, String prevCityName, String curStateCode, String prevStateCode)	Calculates the approximate distance between two transactions.
static Double	gaussianProbability(Double value, Double avg, Double std)	Calculates the gaussian probability (from avg, std) of the given value.
static Double	getMostAccurateLatitude(String zipCode, String cityName, String stateCode)	Calculates the most accurate latitude by zip code, city name and state code.
static Double	getMostAccurateLongitude(String zipCode, String cityName, String stateCode)	Calculates the most accurate longitude by zip code, city name and state code.
static Double	haversine(Double lat1, Double lon1, Double lat2, Double lon2)	Calculates the approximate distance between two locations.

Geo class

Package: `com.feedzai.cloud.udf`

Modifier and Type	Function	Description
Double	easting(lat: Double, lon: Double, granularity: Double = 1.0)	Returns the X ordinate (easting) in kilometers from a latitude and longitude coordinate pair, apart from a granularity scaling factor. The granularity scale factor defaults to 1.0. Scales the result to 1:granularity km.
Double	northing(lat: Double, granularity: Double = 1.0)	Returns the Y ordinate (northing) in kilometers from a latitude coordinate pair, apart from a granularity scaling

Modifier and Type	Function	Description
		factor. The granularity scale factor defaults to 1.0. Scales the result to 1:granularity km.
String	extractContinentFromCountry(country: String)	Returns the continent code correspondent to the input country code.
Double	countryDistInKm(country1: String, country2: String)	Returns the distance of two countries in kilometers.
String	extractCountryCodeFromTopLevelDomain(topLevelDomain: String)	Returns an ISO 3166-1 alpha-2 country code from a top-level domain. Returns "AA" if there is no associated country, and null for invalid top-level domains. Example: <code>Geo.extractCountryCodeFromTopLevelDomain(".pt")</code> returns <code>PT</code> .

Location class

Package: `com.feedzai.pulse.fraud.udf`

Modifier and Type	Function	Description
staticDouble	calculateTravelVelocity(Double distance, Long timestampStart, Long timestampEnd)	Returns travel velocity in kilometers per hour.
staticDouble	distFrom(Double latitude1, Double longitude1, Double latitude2, Double longitude2)	Distance in meters between two points on Earth's surface.
staticDouble	distFromInKm(Double atitude1, Double longitude1, Double latitude2, Double longitude2)	Distance in kilometers between two points on Earth's surface.
static Double	getCartesianX(Double lat, Double lon)	Converts polar coordinates into absolute coordinates.
static Double	getCartesianY(Double lat, Double lon)	Converts polar coordinates into absolute coordinates.
static Double	getCartesianZ(Double lat)	Converts polar coordinates into absolute coordinates.
staticDouble	getHaversineDistance(Double latitude1, Double longitude1, Double latitude2, Double longitude2)	The Haversine distance (distance on a spherical surface) between two points on Earth's surface.
static Double	getLatitudeFromCartesian(Double x, Double y, Double z)	Converts absolute coordinates into polar coordinates.
static Double	getLongitudeFromCartesian(Double x, Double y, Double z)	Converts absolute coordinates into polar coordinates.

LocationBean class

Package: `com.feedzai.common.udfs.distance`

Functions using a JavaBean that stores the location information from the location CSV file in the ProfileUtils class. See the source of the location data in <http://download.geonames.org/export/zip/>.

Modifier and Type	Function	Description
String	getCityName()	Gets the city name of the location record.
Double	getLatitude()	Gets the latitude of the location record.
Double	getLongitude()	Gets the longitude of the location record.
String	getStateCode()	Gets the state code of the location record
String	getZipCode()	Gets the zip code of the location record.
void	setCityName(String cityName)	Sets the city name of the location record.

Modifier and Type	Function	Description
void	setLatitude(Double latitude)	Sets the latitude of the location record.
void	setLongitude(Double longitude)	Sets the longitude of the location record.
void	setStateCode(String stateCode)	Sets the state code of the location record.
void	setZipCode(String zipCode)	Sets the zip code of the location record.
String	toString()	Gets the string representation of the location record.

TaxHeaven class

Package: `com.feedzai.cloud.udf`

Modifier and Type	Function	Description
Boolean	isTaxHaven(countryCode: String)	Returns true if the country is on the taxHavensISO2CountryCodes list of tax haven country codes. Based on Hines' list from 2010. The country code must be an ISO 3166-1 alpha-2 country code.

Zip class

Package: `com.feedzai.pulse.fraud.udf`

Functions for working with postal codes. Country strings must be in [ISO 3166-1 alpha-2](#) format, e.g. US, RU, PT.

Modifier and Type	Function	Description
static Double	distFrom(String country1, String zipCode1, String country2, String zipCode2)	Gets the distance in meters between two zip codes
static Double	getLatitude(String country, String zipCode)	Returns the latitude from a country and zip code pair
static Double	getLongitude(String country, String zipCode)	Returns the longitude from a country and zip code pair
static <code>com.feedzai.pulse.fraud.udf.Zip.ZipData</code>	getZipData(String country, String zipCode)	Returns the <code>{ZipData}</code> object with the specified country and zipCode

Working with business-related data

Card class

Package: `com.feedzai.cloud.udf`

Modifier and Type	Function	Description
Boolean	isValidCardExp(cardExp: String)	Checks if a string is a valid card expiration date using <code>cardExp.matches(regex = "(?:0[1-9] 1[0-2])/[0-9]{2}")</code>
Boolean	isDebit(cardCategory: String)	Check whether the card category is "debit". Case insensitive.
Boolean	isCredit(cardCategory: String)	Check whether the card category is "credit". Case insensitive.
Boolean	isPremium(cardCategory: String)	Check whether the card category is "premium". Case insensitive.
Long	daysUntilExpiration(cardExp: String, timestamp: Long)	Returns the number of days until the card expires, counting from the value received in the timestamp parameter.
Integer	compareNameToCardname(uname: String, cname: String)	Normalizes the two received parameters, and returns the distance between the two strings using <code>Strings.strDistance()</code> . Case insensitive.

Business class

Package: `com.feedzai.cloud.udf`

Modifier and Type	Function	Description
Integer	<code>extractMCGFromMCC(mcc: String)</code>	Returns the merchant category group (MCG) code of a given merchant category code (MCC).

Email class

Package: `com.feedzai.cloud.udf`

Modifier and Type	Function	Description
Boolean	<code>isValid(email: String)</code>	Checks if an email is valid.
String	<code>normalizeEmail(email: String)</code>	Normalizes the given email string by converting to lower case and removing whitespace characters.
String	<code>extractEmailDomain(email: String)</code>	Extracts the domain of a nullable email address.
String	<code>extractEmailTopLevelDomain(email: String)</code>	Extracts the top-level domain of an email address (for example, <code>.pt</code>). Returns null if the email is null or invalid. Example: <code>Email.extractEmailTopLevelDomain("")</code> returns <code>.pt</code> .

EmailUtils class

Package: `com.feedzai.ds.udf`

Modifier and Type	Function	Description
static String	<code>normalizeEmailAddress(String str)</code>	Normalizes an email address: • Removes dots, whitespaces, plus signs, and any tag added after the plus sign. • Converts the email address to lower case. • Converts the <code>googlemail.com</code> domain to <code>gmail.com</code>.

Items class

Package: `com.feedzai.ds.udf`

Modifier and Type	Function	Description
static Boolean	<code>keyContains(Object obj, String key, String value)</code>	Returns true if, for the given key, there is at least one value equal to the provided string value.
static Boolean	<code>keyContains(Object obj, String key, Boolean value)</code>	Returns true if, for the given key, there is at least one value equal to the provided boolean value.
static Boolean	<code>customFieldsContains(Object obj, String key, String value)</code>	Returns true if, for the given key, there is at least one value in <code>custom_fields</code> equal to the provided value.
static Boolean	<code>containsProduct(Object obj, String productName, String productStyle, String productColor)</code>	Returns true if there is at least one product matching the given name, style, and color combination.
static Integer	<code>getDistinctSizes(Object obj)</code>	Returns the number of distinct product sizes in the provided items set, or <code>-1</code> if the input is invalid.
static Integer	<code>getDistinctPrices(Object obj)</code>	Returns the number of distinct prices in the provided items set, or <code>-1</code> if the input is invalid.

Modifier and Type	Function	Description
static Integer	sumQuantity(Object obj)	Returns the sum of item <code>quantity</code> values, or <code>-1</code> if the input is invalid.
static Long	sumPrice(Object obj)	Returns the sum of item <code>price</code> values, or <code>-1</code> if the input is invalid.
static Long	sumPriceUSD(Object obj)	Returns the sum of item <code>price_usd</code> values, or <code>-1</code> if the input is invalid.

Phone class

Package: `com.feedzai.cloud.udf`

Country codes use ISO 3166-1 two-letter country-code format.

Modifier and Type	Function	Description
String	countryCode(number: String)	Returns the country code in ISO 3166-1 two-letter country-code format.
Boolean	isValidForCountry(number: String, country: String)	Returns true if the phone number is valid for a given country. If no country is provided, the function returns false.
String	numberType(number: String, country: String)	Gets the phone number type: Fixed-line, Mobile, Toll-free, Premium Rate, Shared Cost, VoIP and Personal Numbers.
Boolean	isMobile(number: String, country: String)	Returns true if the phone number type is mobile.
Boolean	isTollFree(number: String, country: String)	Returns true if the phone number type is toll-free.
Boolean	isVoIP(number: String, country: String)	Returns true if the phone number type is "VoIP".
Boolean	isPremiumRate(number: String, country: String)	Returns true if the phone number type is Premium Rate ("uan").
Boolean	isSharedCost(number: String, country: String)	Returns true if the phone number type is Shared Cost.
Boolean	isFixedLine(number: String, country: String)	Returns true if the phone number type is Fixed Line.
Boolean	isPersonalNumber(number: String, country: String)	Returns true if the phone number type is Personal Number.
String	normalizePhoneNumber(number: String)	Normalizes a phone number by removing spaces, replacing '+' by '00', and removing all other characters that are not numbers.

Working with timestamps

Time class

Package: `com.feedzai.pulse.fraud.udf`

Modifier and Type	Function	Description
static Long	getDays2Expire(Long timestamp, String expiredate)	Gets the time delta (in days) to expiration.
static Long	getDays2Expire(Long timestamp, String expiredate, String format)	Gets the time delta (in days) to expiration.
static Integer	getHourOfDay(Long timestamp, String timezone)	Gets the hour of the day for a specific timestamp.
	getLocalHourOfDay(Long timestamp)	

Modifier and Type	Function	Description
static Integer		Gets the hour of the day for a specific timestamp assuming the local timezone.
static Integer	getLocalMonth(Long timestamp)	Gets the month for a specific timestamp assuming the local timezone.
static Integer	getLocalMonthDay(Long timestamp)	Gets the day of the month for a specific timestamp assuming the local timezone.
static Integer	getLocalWeekDay(Long timestamp)	Gets the day of the week for a specific timestamp assuming the local timezone. 1 - Monday; 7 - Sunday.
static Integer	getLocalYear(Long timestamp)	Gets the year for a specific timestamp assuming the local timezone.
static Integer	getLocalYearDay(Long timestamp)	Gets the day of the year for a specific timestamp assuming the local timezone.
static Long	getMillis2Expire(Long timestamp, String expiredDate)	Gets the time delta (in milliseconds) to expiration.
static Long	getMillisAtStartOfDay(Integer year, Integer month, Integer dayOfMonth)	Obtains a UTC timestamp (in milliseconds) from the provided calendar year/month/dayOfMonth.
static Integer	getMonth(Long timestamp, String timezone)	Gets the month for a specific timestamp.
static Integer	getMonthDay(Long timestamp, String timezone)	Gets the day of the month for a specific timestamp.
static Integer	getUTCHourOfDay(Long timestamp)	Gets the hour of the day for a specific timestamp assuming the UTC timezone.
static Integer	getUTCMonth(Long timestamp)	Gets the month for a specific timestamp assuming the UTC timezone.
static Integer	getUTCMonthDay(Long timestamp)	Gets the day of the month for a specific timestamp assuming the UTC timezone.
static Integer	getUTCWeekDay(Long timestamp)	Gets the day of the week for a specific timestamp assuming the UTC timezone. 1 - Monday; 7 - Sunday.
static Integer	getUTCYear(Long timestamp)	Gets the year for a specific timestamp assuming the UTC timezone.
static Integer	getUTCYearDay(Long timestamp)	Gets the day of the year for a specific timestamp assuming the UTC timezone.
static Integer	getWeekDay(Long timestamp, String timezone)	Gets the day of the week for a specific timestamp. 1 - Monday; 7 - Sunday.
static Integer	getYear(Long timestamp, String timezone)	Gets the year for a specific timestamp.
static Integer	getYearDay(Long timestamp, String timezone)	Gets the day of the year for a specific timestamp.
static Long	parseDate(String strDate, String format)	Obtains a UTC timestamp (in milliseconds) from a string representation of a date.
static Long	parseExpirationDate(String strDate, String format)	Obtains a UTC timestamp (in milliseconds) from an expiration date string. If the format has no day component, returns the first timestamp of the subsequent month. Example: Time.parseExpirationDate("202003", "yyyyMM") returns the timestamp for 2020-04-01 00:00:00.000 (UTC).
static Long	parseDateOrNullOnError(String strDate, String format)	Checks if the given parameters can be parsed into a Date in milliseconds.

TimeTools class

Package: `com.feedzai.commons.udfs`

Modifier and Type	Function	Description
static Integer	dayOfMonth(Long timestamp)	Returns the "day of month" portion of the given timestamp at the default system timezone.
static Integer	dayOfMonth(Long timestamp, String timezone)	Returns the "day of month" portion of the given timestamp at the given timezone.
static Integer	dayOfWeek(Long timestamp)	Returns the "day of week" portion of the given timestamp at the default system timezone.
static Integer	dayOfWeek(Long timestamp, String timezone)	Returns the "day of week" portion of the given timestamp at the given timezone.
static Integer	dayOfYear(Long timestamp)	Returns the "day of year" portion of the given timestamp at the default system timezone.
static Integer	dayOfYear(Long timestamp, String timezone)	Returns the "day of year" portion of the given timestamp at the given timezone.
static Integer	hour(Long timestamp)	Returns the "hour" portion of the given timestamp at the default system timezone.
static Integer	hour(Long timestamp, String timezone)	Returns the "hour" portion of the given timestamp at the given timezone.
static Integer	minute(Long timestamp)	Returns the "minute" portion of the given timestamp at the default system timezone.
static Integer	minute(Long timestamp, String timezone)	Returns the "minute" portion of the given timestamp at the given timezone.
static Integer	month(Long timestamp)	Returns the "month" portion of the given timestamp at the default system
static Integer	month(Long timestamp, String timezone)	Returns the "month" portion of the given timestamp at the given timezone.
static Integer	second(Long timestamp)	Returns the "second" portion of the given timestamp at the default system timezone.
static Integer	second(Long timestamp, String timezone)	Returns the "second" portion of the given timestamp at the given timezone.
static Integer	weekOfYear(Long timestamp)	Returns the "weekOfYear" portion of the given timestamp at the default system timezone.
static Integer	weekOfYear(Long timestamp, String timezone)	Returns the "weekOfYear" portion of the given timestamp at the given timezone.
static Integer	year(Long timestamp)	Returns the "year" portion of the given timestamp at the default system timezone.
static Integer	year(Long timestamp, String timezone)	Returns the "year" portion of the given timestamp at the given timezone.

MoreTime class

Package: `com.feedzai.ds.udf`

Modifier and Type	Function	Description
static Integer	getMonthLength(Long timestamp, String timezone)	Calculates the number of days of the month for a given timestamp and timezone.

Modifier and Type	Function	Description
static Integer	getUTCMonthLength(Long timestamp)	Calculates the number of days of the month for a given timestamp in UTC timezone.

Working with JSON

Json [class](#)

Package: `com.feedzai.cloud.udf`

For big JSON documents, consider using the functions described in the `AbstractCachedJson` class.

Modifier and Type	Function	Description
String	getString(id: String, json: String, path: String, forceCoersion: Boolean)	Gets a String field from a JSON document, according to a given JSON path expression. If the field is not a String, it can be coerced to one, if the corresponding parameter is set. Otherwise the function logs an error and returns null.
String	getString(id: String, json: String, path: String)	Gets a String field from a JSON document, according to a given JSON path expression. If the field is not a String, the function logs an error and returns null.
String	getAsString(id: String, json: String, path: String)	Gets a field from a JSON document, as a String, according to a given JSON path expression. If the field is not a string, it will be coerced to one.
Boolean	getBoolean(id: String, json: String, path: String)	Gets a Boolean field from a JSON document, according to a given JSON path expression.
Integer	getInteger(id: String, json: String, path: String)	Gets an Integer field from a JSON document, according to a given JSON path expression.
Long	getLong(id: String, json: String, path: String)	Gets a Long field from a JSON document, according to a given JSON path expression.
Float	getFloat(id: String, json: String, path: String)	Gets a Float field from a JSON document, according to a given JSON path expression.
Double	getDouble(id: String, json: String, path: String)	Gets a Double field from a JSON document, according to a given JSON path expression.
String	getAndConcatStrings(id: String, json: String, paths: JList[String], delim: String)	Gets multiple fields, converted to String, from a JSON document as a concatenated string with their values, according to a given JSON path expression.
Boolean	contains(id: String, json: String, path: String)	Checks if a field exists in a JSON document, according to a given JSON path expression.

AbstractCachedJson [class](#)

Package: `com.feedzai.cloud.udf`

This class includes functions for handling big JSON strings. The functions in this class cache the result of the parsing to avoid reparsing on every call. For smaller files, consider using the functions described under the `Json` class.

Modifier and Type	Function	Description
String	getString(id: String, json: String, path: String, forceCoersion: Boolean)	Gets a String field from a JSON document, according to a given JSON path expression. If the field is not a String, it can be coerced to one, if the corresponding parameter is set. Otherwise the function logs an error and returns null.
String		

Modifier and Type	Function	Description
	getString(id: String, json: String, path: String)	Gets a String field from a JSON document, according to a given JSON path expression. If the field is not a String, the function logs an error and returns null.
String	getAsString(id: String, json: String, path: String)	Gets a field from a JSON document, as a String, according to a given JSON path expression. If the field is not a string, it will be coerced to one.
Boolean	getBoolean(id: String, json: String, path: String)	Gets a Boolean field from a JSON document, according to a given JSON path expression.
Integer	getInteger(id: String, json: String, path: String)	Gets an Integer field from a JSON document, according to a given JSON path expression.
Long	getLong(id: String, json: String, path: String)	Gets a Long field from a JSON document, according to a given JSON path expression.
Float	getFloat(id: String, json: String, path: String)	Gets a Float field from a JSON document, according to a given JSON path expression.
Double	getDouble(id: String, json: String, path: String)	Gets a Double field from a JSON document, according to a given JSON path expression.
String	getAndConcatStrings(id: String, json: String, paths: JList[String], delim: String)	Gets multiple fields, converted to String, from a JSON document as a concatenated string with their values, according to a given JSON path expression.
Boolean	contains(id: String, json: String, path: String)	Checks if a field exists in a JSON document, according to a given JSON path expression.

Working with collections and complex fields

Collections class

Package: `com.feedzai.pulse.fraud.udf`

Modifier and Type	Function	Description
static String	getFieldByKey(Object complexField, String key, String keyValue, String fieldToRetrieve)	Retrieves a field from a complex field by matching tuples with the provided key and key value. Returns an empty string on any validation or lookup error. Example: given <code>sender_account_limits</code> with <code>{"event_type": "ACC_LIMIT", "daily_amount": 500000}</code> , calling <code>com.feedzai.pulse.fraud.udf.Collections.getFieldByKey(sender_account_limits, "event_type", "ACC_LIMIT", "daily_amount")</code> returns <code>500000</code> .
static Boolean	contains(java.util.Collection<java.lang.Object> collection, java.lang.Object valueToFind)	Checks if a collection contains a specific value.
static Boolean	containsCaseInsensitive(java.util.Collection<java.lang.Object> collection, java.lang.String valueToFind)	Checks if a collection contains a specific value. Ignores casing when testing for equality.

Collections class

Package: `com.feedzai.cloud.udf`

Modifier and Type	Function	Description
Tuple	getSampleTuple(set: JSet[Tuple])	Retrieves the next tuple from a JSet of Tuples.

CollectionUDFs class

Package: `com.feedzai.pulse.pkernel.runtime`

As the CollectionUDFs class is included in the runtime package of Pulse, you don't need to define an alias to use the functions of this class.



Lists vs Maps

The collection called Map of Pulse Query Language (PQL) and Rule Definition Language (RDL) is the equivalent of a Pulse list created under **Lists** . The following table refers to them as **Maps**.

Modifier and Type	Function	Description
Integer	count(Object collection)	Returns the number of entries in a collection Fails with <i>UnsupportedOperationException</i> if <i>collection</i> is neither a Map nor a Collection.

Modifier and Type	Function	Description
		<pre>// Returns 2 CollectionUDF.count(map("a": 1, "b": 2))</pre>
Boolean	contains(Object collection, Object element)	Checks if <i>element</i> is present in <i>collection</i>. Fails with <i>UnsupportedOperationException</i> if <i>collection</i> is neither a Map nor a Collection. <pre>// Returns true CollectionUDF.contains(map("a": 1), 1)</pre>
Integer	getInt(Object collection, Object? key)	Returns the integer value associated to <i>key</i>. If <i>collection</i> is a List, negative keys start at the end of the list (0 is the first and -1 is the last element). Fails with <i>UnsupportedOperationException</i> if <i>collection</i> is neither a Map nor a List or if the value matching key is not null and not a number. Fails with <i>IllegalArgumentException</i> if <i>collection</i> is a List but <i>key</i> is not a non-null Integer. <pre>// Returns 5 CollectionUDF.getInt(list(1, 3, 5), -1)</pre>
Long	getLong(Object collection, Object key)	Returns the long value associated to <i>key</i>. If <i>collection</i> is a List, negative keys start at the end of the list (0 is the first and -1 is the last element). Fails with <i>UnsupportedOperationException</i> if <i>collection</i> is neither a Map nor a List or if the value matching key is not null and not a number. Fails with <i>IllegalArgumentException</i> if <i>collection</i> is a List but <i>key</i> is not a non-null Integer. <pre>// Returns 3 CollectionUDF.getLong(map("a": 3), "a")</pre>
Float	getFloat(Object collection, Object key)	Returns the float value associated to <i>key</i>. If <i>collection</i> is a List, negative keys start at the end of the list (0 is the first and -1 is the last element). Fails with <i>UnsupportedOperationException</i> if <i>collection</i> is neither a Map nor a List or if the value matching key is not null and not a number. Fails with <i>IllegalArgumentException</i> if <i>collection</i> is a List but <i>key</i> is not a non-null Integer.

Modifier and Type	Function	Description
		<pre>// Returns null CollectionUDF.getFloat(list(null), 0)</pre>
Double	getDouble(Object collection, Object key)	Returns the double value associated to <i>key</i>. If <i>collection</i> is a List, negative keys start at the end of the list (0 is the first and -1 is the last element). Fails with <i>UnsupportedOperationException</i> if <i>collection</i> is neither a Map nor a List or if the value matching key is not null and not a number. Fails with <i>IllegalArgumentException</i> if <i>collection</i> is a List but <i>key</i> is not a non-null Integer. <pre>// Returns 2.5 CollectionUDF.getDouble(list(3.14, 2.5), 1)</pre>
string	getString(Object collection, Object key)	Returns the string value associated to <i>key</i>. If <i>collection</i> is a List, negative keys start at the end of the list (0 is the first and -1 is the last element). Fails with <i>UnsupportedOperationException</i> if <i>collection</i> is neither a Map nor a List or if the value matching key is not null and not a string. Fails with <i>IllegalArgumentException</i> if <i>collection</i> is a List but <i>key</i> is not a non-null Integer. <pre>// Returns "value" CollectionUDF.getString(map("key": "value"), "key")</pre>

Tuples class

Package: `com.feedzai.cloud.udf`

Modifier and Type	Function	Description
AnyRef	getField(tuple: Tuple, field: String)	Returns the field value, or null if the tuple doesn't contain the field or if the tuple is null.

ComplexFields class

Package: `com.feedzai.cloud.udf`

Modifier and Type	Function	Description
String	getString(map: JMap[String, String], key: String)	Get a string field from a string JMap, according to a given key. If the field is not a string or doesn't exist, the function returns null. This function can't be used in Plans, because PQL does not support calls to UDFs with non-primitive parameters. As an alternative use the getStringObj UDF instead.

Modifier and Type	Function	Description
		This function can't be used in Plans , because PQL does not support calls to UDFs with non-primitive parameters. As an alternative use the getStringObj UDF instead.
String	getStringObj(map: AnyRef, key: String)	Get a string field from a string JMap, according to a given key. If the field is not a string or doesn't exist, the function returns null. This function is analogous to getString() but receives the map as an AnyRef. This function is useful when using complex fields in PQL, because PQL does not support calls to UDFs with non-primitive parameters.

Mathematical calculations

Maths class

Package: `com.feedzai.pulse.fraud.udf`

Modifier and Type	Function	Description
static Integer	binNumber(Double value, Double binSize)	Gets the bin number of a value
static Double	coalescedSafeDivision(Double numerator, Double denominator)	Divides numerator by denominator with safeguards for bad divisions
static Double	diffNormalizeAndCap(Double value, Double expected, Double cap)	Computes the difference between a value and the expected, normalized by the expected value, capped to a maximum
static Double	logOfNormalPDF(Double mean, Double std, Double x)	Natural logarithm of normalPDF(Double, Double, Double)
static Double	normalCDF(Double mean, Double std, Double x)	Cumulative distribution function (CDF) for a normal distribution with parameters <code>mean</code> and <code>std</code> at value <code>x</code> .
static Double	minimum(Double a, Double b, Double c)	Three-way minimum
static Integer	minimum(Integer a, Integer b, Integer c)	Three-way minimum
static Double	maximum(Double a, Double b, Double c)	Three-way maximum
static Integer	maximum(Integer a, Integer b, Integer c)	Three-way maximum
static Double	normalPDF(Double mean, Double std, Double x)	Density of the normal distribution with parameters <code>mean</code> and <code>std</code> at the value <code>x</code>
static Double	safeDivision(Double numerator, Double denominator)	Executes a safe division that returns the division value or zero if the operation is not a number or division by zero
static Double	safeDivision(Long numerator, Long denominator)	Executes a safe division that returns the division value or zero if the operation is not a number or division by zero
static Double	toDeg(Double angle)	Converts radians to degrees
static Double	toRad(Double angle)	Converts degrees to radians



PDF

The `normalPDF` function has a corner case defined for when the standard deviation (`std`) is zero. In that case, it will return either 1 if the value `x` equals the mean parameter or 0 otherwise. This might affect your feature distribution because the outcome of `normalPDF` is the likelihood, which is not bound.

Working with model scores

ScoreTransformationUdfs class

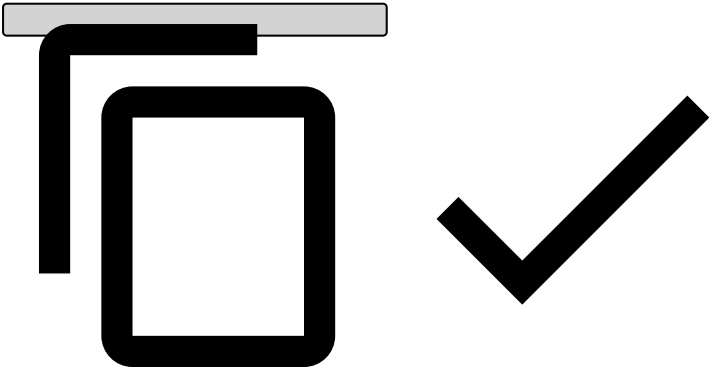
Package: `com.feedzai.ds.udf`

Modifier and Type	Function	Description
static Double	<code>amountDependentScoreTransformation(Double rawScore, Double amount, Double intensity, Double cap)</code>	Transforms the model score using the transaction amount parameterized by an intensity factor and an amount cap.
static Double	<code>thresholdAndAmountDependentScoreTransformation(double rawScore, long amount, double anchor, Object amountBinAndThresholdObject, double defaultThreshold)</code>	Transforms the model using amount-dependent thresholds and an anchor value.

Usage examples:

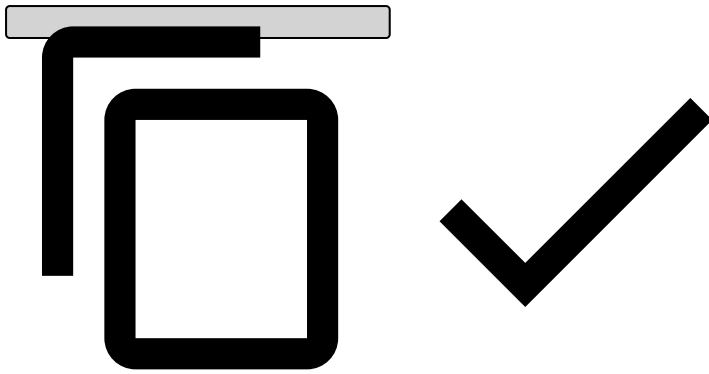
Transforming scores with the amount-dependent score transformation:

```
com.feedzai.ds.udf.ScoreTransformationUdfs.amountDependentScoreTransformation(  
    fraud_true_scored // original score to be transformed  
    amount           // amount field  
    0.3              // lower intensity means lower amounts are greatly affected  
  
    5000000 // when 'amount > cap', the score remains unchanged. In this case scores for amounts  
            over $50k are unchanged  
)
```



Transforming scores with piecewise linear transformations dependent on the amount:

```
com.feedzai.ds.udf.ScoreTransformationUdfs.thresholdAndAmountDependentScoreTransformation(  
    fraud_true_scored, // original score to be transformed  
    amount,           // amount field  
    0.850,            // lower anchor means the raw score is more affected by the transformations  
    {1000: 0.800, 6750: 0.750, 20420: 0.654}, // mapping amounts to thresholds {amount1:  
    threshold1, amount2: threshold2, ...}  
    0.500 // default threshold, ie for amounts greater than the max amount above  
)
```



Cryptography🔒📄

Crypto class🔒📄

Package: `com.feedzai.pulse.fraud.udf`

Modifier and Type	Function	Description
static String	cryptoHash(String algo, Object obj)	Hashes an object using a hashing algorithm.
static String	cryptoHash(String algo, Object obj, String salt)	Hashes an object using a hashing algorithm.
static String	md5(Object obj)	Computes the md5 hash of the object.
static String	sha512(Object obj)	Computes the sha-512 hash of the object.

Debugging🔒📄

Debug class🔒📄

Package: `com.feedzai.pulse.fraud.udf`

Modifier and Type	Function	Description
static Double	debug(Double value, String comment)	Debug function to log a message and a value and return the original object.
static Float	debug(Float value, String comment)	Debug function to log a message and a value and return the original object.
static Integer	debug(Integer value, String comment)	Debug function to log a message and a value and return the original object.
static Long	debug(Long value, String comment)	Debug function to log a message and a value and return the original object.
static Short	debug(Short value, String comment)	Debug function to log a message and a value and return the original object.
static Integer	debugObject(Object value, String comment)	Debug function to log a message and a value.